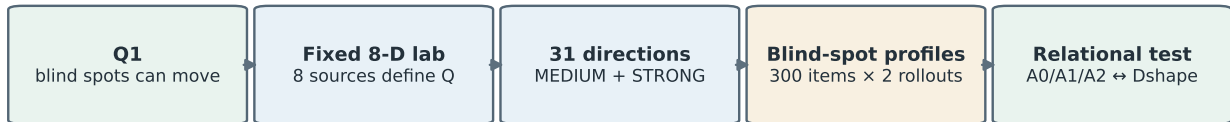
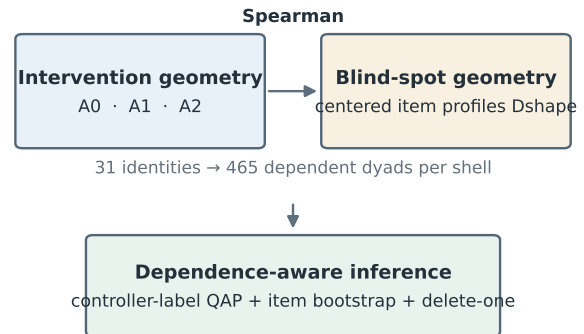
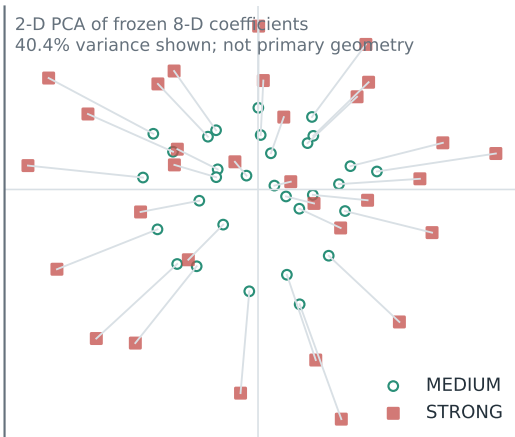


Q2: from causal controllability to relational geometry



Predictors fixed before semantic correctness | Q3 utility remains untested

Intervention laboratory (pre-outcome inset)



Angle and amplitude are separate tested questions — not a manifold theorem